PROJECT PROGRESS REPORT

PROJECT PROGRESS REPORT
(Reporting No. 06 / 10)
Reporting Period: From 20/07/2026 To 27/08/2026

1. Student Information

Student Name	MAN VISHNUBHAI PATEL
Enrollment Number	25IC03CA053
Program/Branch	MCA / ICA
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA (DS/ML) /AI &ML
Mentor Name	Mr. Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Project Title

- OpenCV - Based System for Crack Identification and Width Estimation

4. Progress During the Reporting Period

A. Previous Reporting Review

Planned Task	Status	Remarks
Install & Import Libraries	**Completed**	Required libraries were installed and imported successfully for image processing and crack analysis.
Project Configuration	**Completed**	Project configuration, output directories, and OpenCV parameters were successfully configured.
Output & Mask Directories	**Completed**	Directories for processed images, crack masks, and annotated images were created successfully.
OpenCV Parameters	**Completed**	Gaussian blur, black-hat transformation, morphological processing, and crack-area parameters were configured and tested.
Pixel-to-mm Calibration	**Completed**	Pixel-to-mm calibration parameters were configured and integrated with crack-width estimation; validation using a known reference measurement is being completed.
Image Upload	**Completed**	Image upload functionality was implemented and tested for crack-image analysis.
Image Preprocessing	**Completed**	Image preprocessing was implemented to improve image quality and prepare images for crack analysis.
Crack Segmentation	**Completed**	Crack regions were segmented using image-processing and morphological techniques.
Crack Detection	**Completed**	Crack detection was implemented and tested on input images.
Crack Width Estimation	**Completed**	Crack-width estimation was implemented using detected crack regions and calibration settings.
Risk Classification	**Completed**	Risk-level classification is being developed using measured crack characteristics and defined severity thresholds.
Result Visualization	**In Progress**	Final annotated result visualization is being developed to display detected cracks, measurements, and risk information.

B. Work Completed During This Reporting Period

- **Image Upload:** Implemented and tested image upload functionality for crack analysis.
- **Image Preprocessing:** Completed preprocessing using grayscale conversion, Gaussian filtering and crack enhancement techniques.
- **Crack Segmentation:** Implemented segmentation to separate crack regions from the background.
- **Crack Detection:** Developed and tested the crack detection process using OpenCV.
- **Crack Width Estimation:** Implemented crack-width measurement using detected crack regions and calibration settings.
- **Output Management:** Generated processed images, crack masks and annotated image outputs.
- **Testing:** Tested the developed modules using crack images and reviewed the generated results.

C. Evidence / Deliverables Produced

Sr. No.	Deliverable / Evidence	Description / Link / Attachment
1	Project Source Code	Updated Google Colab source code containing the implemented crack identification and width estimation modules.
2	Image Upload Evidence	Screenshot/output showing successful upload of crack images for analysis.
3	Pre-processed Images	Processed images generated after image preprocessing and crack enhancement.
4	Crack Segmentation Results	Binary crack-mask outputs showing the segmented crack regions.
5	Crack Detection Results	Output images showing the detected crack regions from the input images.
6	Crack Width Estimation	Crack-width measurement results obtained from the detected crack regions.
7	Pixel-to-mm Calibration	Configured calibration parameters for converting pixel-based crack measurements into millimeter.
8	Annotated Images	Output images containing detected cracks and corresponding measurement information.
9	Testing Evidence	Screenshots and results obtained by testing the completed modules with crack images.
10	Project Progress Evidence	Updated project structure showing Modules 1–8 completed, with Risk Classification, Result Visualization, CSV Report, and ZIP Download remaining for further development.

Project Details:

```
Project Name       : OpenCV-Based System for Crack Identification and Width Estimation
Maximum Image Size : 1600
Gaussian Kernel    : 5
CLAHE Clip Limit   : 2.0
BlackHat Kernel    : 21
Morphology Kernel  : 3
Minimum Crack Area : 20
Measurement Step   : 20
Calibration        : ENABLED
Pixels per MM      : 10.0
MM per Pixel       : 0.1
GOOD threshold     : < 0.3 mm
MEDIUM threshold   : 0.3 - 1.0 mm
HEAVY threshold    : > 1.0 mm
Project Configuration completed successfully.
Output directories successfully.
Image processing parameters successfully.
Calibration successfully.
Severity thresholds successfully.
Visualization successfully.
```

Image Upload Functions:

```
Please select one or multiple crack images.
Choose Files 201.jpg
201.jpg(image/jpeg) - 26980 bytes, last modified: 5/10/2017 - 100% done
Saving 201.jpg to 201.jpg
 Accepted image: 201.jpg
------------------------------------------------------------
Total images uploaded: 1
------------------------------------------------------------
1. 201.jpg
 Image upload successfully.
```

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Processing Image Functions:

```
============================================================
PROCESSING IMAGE 1/1: 201.jpg
============================================================

------------------------------------------------------------
PROCESSING: 201.jpg
------------------------------------------------------------
 Image read: 480 x 320
 Resize: 480 x 320
 Otsu threshold value: 30.00
 Crack regions detected: 50
 Skeleton pixels: 10520
 Area: 75485.000 pixels²
 Length: 12826.266 pixels
 Maximum Width: 2.718 mm
 Average Width: 0.702 mm
 Severity: HEAVY
 Risk Level: HIGH
 Health Message: Crack width is greater than 1.00 mm. The detected crack is classified as high risk.
 201.jpg processed successfully.

============================================================
MULTI-IMAGE PROCESSING COMPLETED
============================================================
 Uploaded       : 1
 Successful     : 1
 Failed         : 0
completed successfully.
```

D. Current Project Progress

Overall Project Completion: 70 %

Sr. No.	Project Component	Status	Completion
1	Library Installation	Completed	100%
2	Import Libraries	Completed	100%
3	Project Configuration	Completed	100%
4	Image Upload	Completed	100%
5	Image Preprocessing	Completed	100%
6	Crack Segmentation	Completed	100%
7	Crack Detection	Completed	100%
8	Crack Width Estimation	Completed	100%
9	Risk Classification	In Progress	50%
10	Result Visualization	In Progress	50%
11	CSV Report	Pending	0%
12	ZIP Download	Pending	0%

E. Challenges / Issues Faced

- Variations in image quality, lighting, shadows, surface texture, and camera conditions affected crack detection.
- Selecting suitable image-processing techniques and OpenCV parameters required research and experimentation.
- Different crack sizes, shapes, orientations, and image resolutions made consistent analysis challenging.

- Accurate crack segmentation, thin-crack detection, width estimation, and pixel-to-mm calibration required parameter tuning.
- Multiple test images were required to identify errors and improve detection and measurement reliability.
- Integrating detection, measurement, risk classification, visualization, and reporting into one workflow required additional testing and development.

F. Actions Taken / Solution Implemented

- Applied Gaussian filtering to reduce image noise.
- Used black-hat transformation to enhance dark crack regions.
- Applied morphological operations to improve crack segmentation.
- Used minimum crack-area filtering to remove unwanted small regions.
- Implemented crack detection based on the segmented crack regions.
- Implemented crack-width estimation using the detected crack information.
- Configured pixel-to-mm calibration for physical crack-width measurement.
- Tested the developed modules with multiple crack images.

G. Plan for Next Reporting

Sr. No.	Planned Activity	Expected Deliverable	Target Date
1	Complete Risk Classification	Low, Medium and High-risk classification results	03 September 2026

2	Complete Result Visualization	Annotated images with crack detection, length, width and risk information	03 September 2026
3	CSV Report generation	Structured CSV file containing crack analysis and measurement results	10 September 2026
4	ZIP Download functionality	Downloadable ZIP containing images, masks and reports	10 September 2026
5	Perform complete system testing and error handling	Tested and stable crack-analysis system	17 September 2026
6	Optimize parameters and complete final validation	Improved detection accuracy and validated results	17 September 2026
7	Complete project documentation and final demonstration	Final project report, presentation and demonstration-ready system	24 September 2026

H. Learning / Skills Acquired

- Learned image pre-processing, crack segmentation, crack detection, skeletonization, crack length and width estimation, and pixel-to-mm calibration.
- Gained practical experience with Python, OpenCV, NumPy, Pandas, Matplotlib, Scikit-image, and Google Colab.
- Developed research skills through literature review and analysis of existing crack detection techniques.
- Improved problem-solving skills by handling image noise, lighting variations, thin cracks, segmentation errors, and measurement issues.
- Gained testing skills by testing image-processing modules with multiple crack images and reviewing results.

- Improved documentation skills through technical reporting, result recording, and project documentation.
- Developed teamwork and communication skills through progress discussions, technical explanations, and project presentations.

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4. Tasks Accomplished

- Completed installation and import of all required Python and image-processing libraries.
- Completed project configuration and created output directories for processed images, crack masks, and annotated images.
- Implemented image upload and image preprocessing functionality.
- Implemented crack segmentation and crack detection using OpenCV-based techniques.
- Implemented crack width estimation with pixel-to-mm calibration configuration.
- Tested the developed modules using multiple crack images and reviewed the generated outputs.
- Prepared the system for risk classification, result visualization, CSV reporting, and ZIP download.

5. Tasks in Progress / Next Plan

- Complete the risk classification module using crack-width measurements and defined severity thresholds.
- Complete result visualization with detected cracks, crack length, crack width, and risk status.
- Develop CSV report generation for storing image-wise crack analysis results.
- Implement ZIP download functionality for annotated images, crack masks, processed images, and CSV reports.
- Perform complete system testing using multiple crack images and identify remaining errors.

- Optimize OpenCV parameters and improve crack detection and measurement accuracy.
- Complete final validation, documentation, project integration, and demonstration preparation.

6. Learnings / Skills Acquired

- Gained practical knowledge of OpenCV-based image processing and automated crack analysis.
- Learned techniques for crack preprocessing, enhancement, segmentation, and detection.
- Developed understanding of crack skeletonization, length measurement, and width estimation.
- Learned pixel-to-mm calibration for converting image measurements into real-world dimensions.
- Improved Python programming skills using NumPy, Pandas, Matplotlib, and Scikit-image.
- Gained experience in Google Collab, multiple-image processing, output management, and result analysis.
- Improved problem-solving, testing, debugging, and validation skills during system development.
- Developed technical documentation and research skills for presenting project methodology and results.

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References:

- **Ramasamy, L., & Vijayan, P. (2026).** *Advanced Image Processing and Deep Learning Methods for Automated Crack Detection in Civil Engineering Structures: A Comprehensive Review.* Asian Journal of Civil Engineering.
- **Li, Y. (2026).** *Crack Detection and Semantic Segmentation in Concrete Structures Using an Improved Deep Learning-Based Framework.* Journal of Engineering and Applied Science, 73, 107
- **Grzesiak, S., Nogatz, T., de Sousa, C., Kronenberger, M., Redenbach, C., & Pahn, M. (2026).** *In Situ 4D Image Analysis of Crack Width and Strain Changes in GFRP-Reinforced Concrete.* Materials and Structures, 59, 197.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks
Progress as per Previous Plan	L3– Apply	All planned tasks completed	Most completed	Partial completion	Very little progress	
Technical / Research Implementation	L4– Analyze	Excellent implementation with clear evidence	Good implementation	Basic implementation	Insufficient evidence	
Quality of Deliverables	L5– Evaluate	High-quality, validated deliverables	Good quality	Acceptable	Poor / incomplete	
Problem Solving & Challenges	L4– Analyze	Challenges clearly identified and effectively solved	Most issues addressed	Limited solutions	No clear effort	
Planning for Next Phase	L6– Create	Clear, realistic and measurable plan	Mostly clear plan	Basic plan	No proper plan	
Documentation & Presentation	L3– Apply	Well-structured and professional	Minor issues	Average clarity	Poor documentation	
Total marks out of 100						

Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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